Brainstorming and Idea Prioritization

Phase 1: Brainstorming & Ideation

1. Purpose of Brainstorming

The purpose of brainstorming was to choose a practical problem that a web application and generative AI can address within a student-project scope. Travel planning was selected because a traveller has to combine destination, dates, budget, group size, pace, food needs and interests before a usable plan exists. These details are usually collected from many websites and arranged by hand into a day-wise schedule.

Each idea was judged on feasibility as well as novelty: it needed a clear user journey, a visible result that can be demonstrated in a browser, and a design that can be explained from source code. trvale-book offers a defined input-output workflow in which an AI model, a database and a budget calculator work together.

2. Ideas Considered

AI-powered itinerary planner with budget tracking

This idea turns a structured set of preferences into a day-by-day plan with morning, afternoon and evening activities, then lets the user review, edit, recalculate the budget and print the result. It was prioritized because it is useful as a planning assistant without needing booking or payment features.

General destination information portal

A portal of attraction descriptions would still leave the user to assemble a schedule. Without personalization it does not use the strengths of a generative model, so it was ranked lower.

Complete booking platform

Booking transport, hotels and restaurants needs provider contracts, verified inventory and payment security. These are beyond the current application and remain future scope; they are not claimed as implemented.

Standalone expense tracker

A tracker records spending but does not create the plan. Budget awareness was instead built into the planner: the app compares the target budget with the estimated cost of the generated schedule.

3. Prioritization Criteria

- User value: reduce the repetitive effort of building a first itinerary draft.
- Feasibility: achievable with a React frontend, a FastAPI backend and a hosted database.
- AI relevance: the model produces meaningful, structured content rather than decoration.
- Demonstrability: the whole flow can be shown on the deployed site, even without any API keys (demo mode).
- Extensibility: authentication, live place data and booking can be added later.

4. Selected Idea and Rationale

trvale-book was selected as the strongest fit. A five-step wizard collects the inputs; the backend validates them, requests a structured itinerary from Google Gemini or OpenAI (or uses a built-in heuristic generator when no key is configured), and calculates a budget breakdown. Trips can be saved, searched, edited, deleted and printed. Together these form one coherent product rather than a set of unrelated AI features.

5. Scope Guardrails

The brainstorming outcome separates what exists from what is only planned. The application creates an itinerary draft with estimated costs. It does not book services, take payments, or verify hotel prices, opening hours, ratings or availability. Estimated costs are model or heuristic output and must be checked by the traveller.